
Python Basics Cheatsheet

Python for Data Processing

1. Basic Calculation Operations

```
5 + 3 # 8           5 / 3 # 1.66...
5 - 3 # 2           5 // 3 # 1
5 * 3 # 15          5 % 3 # 2
5 ** 3 # 125
```

```
# Rounding
round(1.6667) # 2
round(1.6667, 2) # 1.67
int(1.6667) # 1
```

2. Basic Type Conversions

```
int("42") # String to int
float("3.14") # String to float
str(42) # Int to string
list("abcd") # String to list
```

3. Lists

```
# Create a list
my_list = [3, 2, 1]

# Operations
item = my_list[2] # 1 - element at index 2
my_list.append(3) # [3, 2, 1, 3]
new_list = my_list + [5, 6] # [3, 2, 1, 3, 5, 6]
x = len(new_list) # 6 - length of list
"Apple" in new_list # False
```

```
# Looping 1
for item in my_list:
    print(item)
```

```
# Looping 2
for i in range(len(my_list)):
    item = my_list[i]
    print(item)
```

4. List methods and slicing

```
# Methods
new_list.sort() # [1, 2, 3, 3, 5, 6]
new_list.insert(2, 10) # [1, 2, 10, 3, 3, 5, 6]
new_list.remove(3) # [1, 2, 10, 3, 5, 6]
i = new_list.index(10) # 2 (↑ index of 10: 2)

# Slicing
lst = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
lst[2:5] # [2, 3, 4] (from 2 to 4)
lst[:3] # [0, 1, 2] (from start to 2)
lst[7:] # [7, 8, 9] (from 7 to end)
lst[-3:] # [7, 8, 9] (last 3)
lst[::-2] # [0, 2, 4, 6, 8] (every 2nd)
lst[1:8:3] # [1, 4, 7] (from 1 to 7 step 3)
lst[::-1] # [9, 8, ...] (reverses list)
lst[8:2:-2] # [8, 6, 4] (from 8 to 3 step -2)
```

5. Strings

```
# Strings
my_str = "Hello, World!"

my_str.lower() # Convert to lowercase
my_str.upper() # Convert to uppercase
my_str.capitalize() # Only first char uppercase
my_str.lstrip() # Remove leading whitespace
my_str.rstrip() # Remove trailing whitespace
my_str.strip() # Combined lstrip and rstrip
my_str.isalpha() # all characters alphabetic?
my_str.isdigit() # all characters digits?
my_str.islower() # all characters lowercase?
my_str.isupper() # all characters uppercase?
my_str.split(" ") # split string on spaces
```

```
# Looping through characters 1
for char in my_str:
    print(char)
```

```
# Looping through characters 2
for i in range(len(my_str)):
    char = my_str[i]
    print(char)
```

6. Dictionaries

```
# Create a dictionary
my_dict = {'animal': 'monkey', 'age': 43}

# Operations
animal_type = my_dict['animal'] # 'monkey'
my_dict['name'] = 'King Julian' # adding to dict
x = len(my_dict) # 3
'age' in my_dict # True
43 in my_dict # False
```

```
# Looping over keys and getting values
for key in my_dict:
    print(key)
    print(my_dict[key])
```

7. Reading files

```
# Open and read whole file
file = open("file.txt", "r")
content = file.read()
print(content)
file.close()
```

```
# Open and read line by line
file = open("file.txt", "r")
for line in file:
    print(line)
file.close()
```

```
# Alternative method: with open
with open("file.txt", "r") as file:
    for line in file:
        print(line)
```
